



Enzymes to recycle plastic

A French company is using enzymes to recycle one of the most common single-use plastics. <https://digit.in/nov21-17>



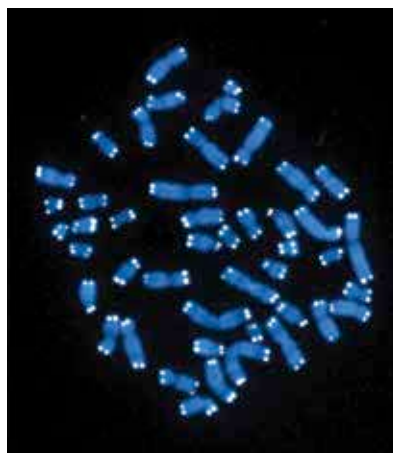
Goodbye transistors?

New optical switches offering up to 1,000x better performance over transistors. <https://digit.in/nov21-18>

experiment indicated a fundamental mechanism common to all life as we know it. It is now known that the telomeres with their characteristic sequence are found in most life forms, from amoeba to plants to animals. Greider, a student of Blackburn discovered an enzyme, telomerase, that promotes the formation of telomeres, slowing down ageing. If there are long telomerase sequences, then the senescence in the cell is reduced. Both Blackburn and Szostak observed that as the organism ages, mutations cause the telomerase sequences to become shorter and shorter. The discovery of telomerase was made on Christmas Day, 1984. Scientists have speculated that telomere length all over the body may be fundamentally linked to ageing, and therapies that can preserve telomere length are being investigated.

This is known as telomere attrition, one of the 13 hallmarks of ageing identified by a group of researchers in 2013. Another is an accumulation of somatic mutations, mutations of the DNA within an organism that generally does not get passed on to the next generation (the near immortal Hydra, that procreates by budding is a rare exception to the rule). Somatic mutations falls inside the bucket of genomic instability, which is any kind of deterioration in the genetic information. Protein homeostasis is a degradation in the ability of the body to fold proteins properly. These damaged proteins have been observed to increase in the brain with age. Though there are molecular chaperones that assist in the process of the folding, and make sure that the proteins turn out right, with ageing, these chaperones are overwhelmed. There are hints that mitochondrial degradation may play a major role, but the organelle is difficult to investigate. Apart from mitochondrial dysfunction, there is protein damage, oxidative stress, or altered intercellular communication, also known as dysregulation of cellular signalling. Altered intercellular communication

covers a range of agents, including insulin and cytokines. Just one of the many processes covered under altered intercellular communication is DNA methylation, which is the addition of the methyl group to typically the C base in a DNA strand, although methylation has been observed with A too. The number of methylated C bases increases with age, it is a straightforward relationship, but the exact molecular processes are still little understood. In fact, the extent of the methylation in the DNA can reveal the epigenetic age of the organism, a genetic clock



Human chromosomes with telomeres highlighted in white

for senescence. The more methylated the DNA in an organism is, the closer it is to kicking the bucket. Computer models are used for methylation analysis, and can accurately predict the methylation in an organism.

Scientists discovered that there is not one molecular mechanism of aging but many. Here is where things get complex beyond human comprehension. All of these different mechanisms of aging interact with each other in the body. There are some signals from stress, some from the diet, some from the environment, that are all processed on a cellular level by various tissues and organs in different ways. It is possible, for example, for your skin to be older than your heart, if you have too much sun exposure, but have been other-

wise eating, sleeping and exercising regularly. A lot of the research is conducted on the network itself, how all these molecular mechanisms interact with each other, as against what each of them do individually. Since the 1990s, researchers began making mathematical models for how all these different biological processes interact with each other, and since the 2000s, computers have joined the investigation.

The quest for the fountain of youth is age old, but the prospects for finding it are better now since it is scientists who are on the hunt, not adventurers, alchemists or apothecaries. A team of researchers have successfully extended the life spans of mice by 50 percent by boosting the amount of telomerase in the body. The problem with this therapy is that boosting the amount of telomerase also encourages the growth of cancerous cells. Biohackers are experimenting with a range of supplements that can potentially increase the lifespans of humans, and pharmaceutical companies around the world are actively developing anti-aging drugs. One of the approaches used by these drugs is to simulate a calorie restriction diet, tricking the body into thinking that it is receiving less food than it actually is. This line of research is inspired by a series of studies on everything from flies, to rats, to mice, to humans, where restricting the amount of food intake, somewhere between starvation and satiation, is the only diet proven to extend lifespans.

There are over 300 theories on what causes aging, but all of them fall into two categories, damage accumulation where multiple systems in the body degrade over time, and the other one is programmed aging theories. The latter are a cluster of theories which investigate the possibility that maybe in the environment that humans have evolved, either the body is programmed to get old, or simply does not have a gameplan in place to tackle such a long life. 

Credit: Hsiao-Padilla-Nash/Thomas Ried/National Cancer Institute/National Institutes of Health